

PatchWise Agent2Agent (A2A) Review

Aim: convert patch review from manual babysitting to autonomous, evidence-backed rounds with strict LGTM safety gates and resumable execution.

Builder: Chanakya

Reviewer: Aryabhata

Driver: a2a CLI Orchestrator

What A2A Solves

- Stops repeated restart-from-round-1 cycles.
- Maintains prior findings and round context.
- Prevents silent/unsafe LGTM decisions.
- Creates audit trail for every decision.

Tasks Covered

- Patch/lore intake and round orchestration.
- Builder fix generation and patch updates.
- Reviewer findings with severity and evidence.

- Validation gates (schema/checkpatch/static).

- Auto-respin (vN -> vN+1) and reports.

Building Blocks

- Session manager (`.a2a/sessions`).

- Artifact engine (`.a2a/reports`, `.a2a/logs`).

- Findings lifecycle (open/new/resolved/closed).

- Worktree lock and concurrency guard.

- Lore ingestion + prior comment mapping.

- Email bridge (optional remote operation).

Call Flow

1. Input Local patch path, lore URL/msgid, or existing session.

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2. Session Init Load config, task, round budget, prior context.

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3. Builder Round Apply fixes, explain rationale, publish residual risks.

4. Validation Gate Schema + optional checkpatch/static checks.

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5. Reviewer Round Emit structured findings and verdict.

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6. LGTM GateBlock if open findings or inconsistent reviewer reasoning.

7A. PassLGTM -> summary + HTML report + optional next-version patch.

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7B. FailInject findings into next round and continue.

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8. Stop RulesMax rounds/time budget/operator stop, then resumable.

Execution Commands

a2a loop --task <name> --watch-path /abs/path --max-rounds 3

a2a loop --task <name> --lore-url "https://lore.kernel.org/all/<msgid>/"

a2a loop --session <session-id>

./run.sh <lore-url|msgid|session|path>

Time Save (Estimated)

Workflow	Manual	A2A	Saving
3-round patch cycle	120-180 min	40-70 min	45-65%
Context rebuild after interruption	20-40 min	3-8 min	70-90%
Report + evidence packaging	25-45 min	5-12 min	65-85%

Indicative range based on internal operator workflow observations; actual values depend on patch complexity and tool latency.

Achievements

- Autonomous multi-round loop with resume.
- Findings-gated LGTM enforcement.
- Session HTML report auto-generation.
- Lore-driven patch review and respin path.
- Prior-comments + subsystem scan support.

Possible Use Cases

- Kernel patch pre-submit quality gate.
- Lore review triage and suggested replies.
- Team handoff: resume from session id.
- Remote review control via email bridge.
- Patchset consistency checks across versions.